Decision and Risk Analysis

Introduction

DAEN 427/ISEN 427/ISEN 627

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Fall 2026

What have I gotten myself into?

DAEN/ISEN 427/627 is a rigorous introduction to decision and risk analysis. The course:

- **Introduces core Bayesian statistical methods** used in data science and engineering research;
- **Emphasizes understanding**, data-driven decision-making, clear communication of results, and critical evaluation of evidence;
- **Draws on relevant examples** from engineering, psychology, economics, and public health policy; and
- **Uses modern software** to reproducibly analyze and interpret data in support of sound scientific conclusions.

Decision and Risk Analysis

Good decisions require four questions:

Choices $\rightarrow$ **Outcomes** $\rightarrow$ **Uncertainty** $\rightarrow$ **Preferences**

Decision and Risk Analysis

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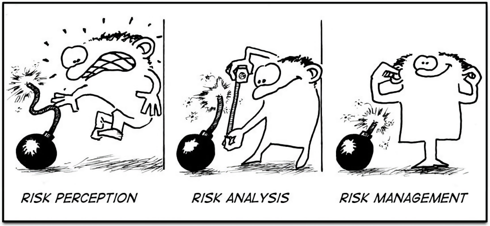
1. **What can we do?**
   Identify our possible choices.
2. **What could happen?**
   Think about possible outcomes and probabilities.
3. **What if we are unsure?**
   Make decisions even when information is incomplete.
4. **What matters most?**
   Compare outcomes, trade-offs, and preferences.

500 kg of US dollars or 500 kg of gold.

What would you choose? What would someone else choose?

Are we modeling risk itself,
or merely the aspects of uncertainty that are easiest to quantify?



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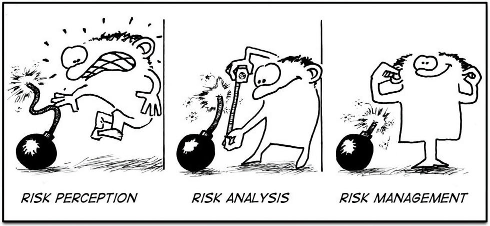

Example: A flood risk model might estimate: "There is a 2% annual probability of flooding, with an expected property loss of $40,000." That is mathematically convenient: *probability x expected loss*. But it may ignore whether residents trust the model or can afford insurance, or political leaders delay action, or infrastructure fails under stress. So the question becomes: **"Are we modeling the full risk of flooding, or just the measurable part: water levels, probabilities, and dollar losses?"**

Alexander Abuabara, PhD

- About me: `https://abuabara.github.io`
- **To be a great teacher ...**

*Yo, I'll tell you what I want, what I really really want**

- **To be a great teacher ...** that means always learning, adapting, and getting better each and every day.
- **That you learn** concepts, analyze data, pass the class ...
  ... and go do great things in life!

* Wannabe by Spice Girls

- At first, it may seem that answers are the most important.
- But the real goal is **to learn how to ASK questions**.

- When you can create thoughtful questions about a topic means you truly understand it.
- **Syllabus**!